Controlled Hyperparameter and Model Experiments

Project: AI-Enabled Underwater Garbage Detection Using Computer Vision
Dataset: TrashCAN – material_version
Experiment period: V1, V2 and V2.1 experiments

1. Objective

The objective was to evaluate different YOLO configurations, investigate the effect of training parameters and data-sampling strategies, and identify a suitable model for underwater garbage detection.

The experiments focused on:

- Comparing YOLO model variants.
- Evaluating different image resolutions.
- Investigating learning-rate changes.
- Studying underwater-specific augmentation.
- Addressing class imbalance through oversampling.
- Comparing performance using validation metrics.

2. Dataset versions

Version	Images	Important characteristics
V1	7,212	Initial annotated dataset
V2	7,212	Exploratory version; video overlap discovered
V2.1	7,212	Revised video-level split with no video overlap

V2.1 contained 5,078 training images, 1,050 validation images and 1,084 test images, distributed across 312 source videos.

The V2.1 split was designed to prevent frames from the same source video appearing in multiple subsets.

3. V1 model experiments

Experiment	Model/configuration	Precision	Recall	mAP@50	mAP@50–95
E1	YOLO11n, 640px	0.249–0.252	0.188	0.186	0.119

E2	YOLO11s, 640px	0.394	0.198	0.185	0.125
E3	YOLO11s, LR 0.001	0.316	0.169	0.151	0.0984
E4	YOLO11s, 960px	0.210	0.252	0.191	0.120
E5	YOLO11s, class-balanced oversampling	0.302	0.245	0.219	0.143
E6	YOLO11s, underwater augmentation	0.306	0.228	0.200	0.1426

The results showed that changing the image resolution affected the precision-recall balance. Targeted oversampling improved aggregate mAP compared with the initial V1 YOLO11s baseline, although several rare classes remained difficult to detect.

4. V2 exploratory experiment

An exploratory model was trained using the V2 dataset and produced promising results. However, subsequent dataset auditing revealed overlap between source videos across the training, validation and test splits.

Therefore, V2 results were not treated as directly comparable with the leakage-safe V2.1 experiments. Exact V2 metrics should be taken from the original training log before numerical reporting.

5. V2.1 experiments

All three experiments below used 640 × 640 input resolution and 100 training epochs.

Experiment	Model	Precision	Recall	mAP@50	mAP@50–95
E7	YOLO11s baseline	**0.519**	0.234	**0.236**	**0.143**
E8	YOLO11s oversampled	0.494	0.198	0.203	0.125
E9	YOLOv8s baseline	0.439	**0.238**	0.216	0.130

6. Observations

- YOLO11s baseline achieved the highest aggregate validation precision and mAP among the three V2.1 experiments.
- Oversampling did not improve overall validation performance.
- YOLOv8s achieved slightly higher recall than YOLO11s but lower precision and mAP.
- Increasing image resolution in V1 improved recall but did not improve mAP@50–95.

- Learning-rate changes and underwater augmentation produced different precision-recall trade-offs.

7. Selected configuration

The V2.1 YOLO11s baseline was selected for subsequent evaluation.

Parameter	Selected value
Architecture	YOLO11s
Initial weights	Pretrained
Dataset	V2.1
Image size	640 × 640
Epochs	100
Parameters	9.42 million
GFLOPs	21.5
Validation precision	0.519
Validation recall	0.234
Validation mAP@50	0.236
Validation mAP@50–95	0.143

8. Conclusion

The experiments identified YOLO11s baseline on the leakage-safe V2.1 dataset as the strongest completed configuration based on aggregate validation performance. The selected candidate was retained for further evaluation, while the final test set remained separate from model selection.

Note: The recorded results are validation metrics. They should not be described as final test performance. Exact training settings beyond those listed should be recovered from the corresponding training configuration files.